

Nicholas Johnson

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Portfolio

EDUCATION

George Mason University

Expected Graduation: Summer 2027

B.S. Neuroscience, Concentration: Computational Neuroscience

Fairfax, VA

- Curriculum focused on Molecular and Cellular Neuroscience, Computational Neuroscience, Neural Networks, and Neural Informatics
- Treasurer, Students in Neuroscience — Administering the Fall 2026 budget and overseeing expense tracking for the student organization
- Selected as one of 18 from 60 applicants for the GEO Neurotechnology in Germany Program — 6-credit intensive visiting 8 research labs across Dortmund, Mannheim, Cologne, Munich, Augsburg, and Berlin (*Summer 2026*).

Northern Virginia Community College

2023 – 2025

A.S. Engineering

Annandale Campus, VA

- Curriculum focused on general education and core STEM

Warren County High School

Front Royal, VA

PROJECTS

Hodgkin–Huxley Computational Neuroscience Project

Independent Research Project | May 2026 – Present

- Designed and implemented a biophysical neuron simulator from the mathematical framework of the Hodgkin–Huxley model to model action potential generation and membrane dynamics
- Investigated how sodium and potassium conductance parameters alter neuronal spike morphology and firing behavior through computational experimentation
- Validated simulation outputs against NEURON to evaluate numerical accuracy and biological realism
- Interpreted the biological significance of conductance-dependent firing patterns and proposed future extensions for computational neuroscience applications
- Authored an independent research-style scientific paper documenting methodology, results, analysis, and future directions

RESEARCH

Undergraduate Research Assistant

January 2026 – Present

George Mason University | Under the supervision of Dr. Herin

Fairfax, VA

- Collaborating on manuscript preparation for a comparative anatomy and physiology education study
- Contributing to data interpretation, scientific writing, editing, and narrative development for peer-reviewed publication

WORK EXPERIENCE

MRI Office Assistant

July 2025 – Present

Institute for Biohealth Innovation, George Mason University

Fairfax, VA

- Completed administrative tasks as requested by research administration staff
- Received entry-level MRI training from a certified MRI Technologist

Office Assistant, IET Department

Feb 2025 – May 2025

Northern Virginia Community College

Annandale Campus, VA

- Created simulation labs in computational computer networks
- Debugged and transferred hardware in student-rental laptops; configured networks and switches for IT labs
- Completed general office and lab duties as assigned